COMPETENCY STANDARD

FOR

JACQUARD PROGRAMMING

(RMG SECTOR)

Skills for Industry Competitiveness and Innovation Program (SICIP)
Finance Division, Ministry of Finance

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INTRODUCTION

The Skills for Industry Competitiveness and Innovation Program (SICIP) has the overall objective of developing a skilled workforce adept at handling new technologies, especially for emerging industries in Bangladesh. It will expand skills training and strengthen the development of the training ecosystem to address the skills requirements of the SICIP-selected industry sectors. The program aims to (i) increase the technology-oriented skilled workforce across emerging and priority sectors, (ii) promote inclusive skilling and upskilling opportunities for women and socially disadvantaged groups, (iii) incentivize industry-university partnerships to nurture innovation capacity and improve industry competitiveness, and (iv) foster skills for climate-resilient manufacturing processes and green technologies. The program is expected to benefit about 220,000 new and existing workers over a 6-year implementation period from 2024-2029.

The SICIP Program has, therefore, taken the initiative to enhance the employability and productivity of trainees by implementing market-responsive and job-focused training programs through public and private training providers. This will require the development of competency standards for each of the occupations/trades which will provide a structured framework in the learning process to guide training providers, ensure consistent training quality, and create an alignment between the skills provided by the training institutes and the needs of the industry.

This competency standard is therefore developed to improve skills following the job roles and skill sets of the occupation and ensure that the required skills are aligned with industry requirements.

The Competency Standard also suggests integration of YouTube or similar digital platforms or downloaded clips into classroom practice to ensure simulated creation of the contents so that learners are exposed to visual demonstrations before classroom instruction or practical session, which aligns with modern learning preference and supports flipped classroom models.

The document details the format, sequencing, wording, and layout of the Competency Standard for an occupation which comprises Units of Competence and its corresponding Elements.

OVERVIEW:

A **Competency Standard** is a written specification of the knowledge, skills, and attitudes required for the performance of a job or occupation or trade corresponding to the standard of performance required in the workplace.

Competency standard:

- provides a consistent and reliable set of components for training, recognizing, and assessing people's skills, and may also have optional support materials.
- enables industry-recognized qualifications to be awarded through direct assessment of workplace competencies
- encourages the development and delivery of flexible training that suits individual and industry requirements
- encourages learning and assessment in a work-related environment which leads to verifiable workplace outcomes.

Competency Standard has been developed by a working group comprised of occupation-specific experts from the industry/institution and relevant consultants of SICIP.

Competency Standards describe the skills, knowledge, and attitude needed to perform effectively in the workplace. Competency Standards acknowledge that people can achieve vocational and technical competency in many ways by emphasizing what the learner can do, not how or where they learned to do it.

With Competency Standards, assessment and training may be conducted at the workplace, at training organization, during regular work, or through work experience, work placement, work simulation or any combination of these.

A Unit of Competency describes a distinct work activity that would normally be undertaken by one person in accordance with industry standards.

Units of Competency are documented in a standard format that comprises:

- Reference to Industry Sector, Occupational Title and Occupational Description
- Unit code
- Unit title
- Unit descriptor
- Unit of Competency
- Elements and performance criteria
- Variables and range statement
- Evidence guides

Together all the parts of a Unit of Competency:

- Describe a work activity
- Guide the assessor in determining whether the candidate is competent.

Identification and validation of units of competency and elements for each occupation were made by experts from various Light Engineering companies in industry consultative workshops.

The ensuing sections of this document comprise a description of the respective occupation with all the key components of a Unit of Competence:

- An overview of all Units of Competence for the occupation and their corresponding duration required for completion of training.
- The Competency Standards that include the Unit of Competency, Unit Descriptor, Elements and Performance Criteria, Range of Variables, Curricular Content Guide, and Assessment Evidence Guide.

Units & Elements at a Glance:

Generic Competencies (30 hrs.)

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP-RMG-JP-01-G	Apply OHS Practices in the Workplace	1. Identify, Control and Report OHS Hazards 2. Conduct Work Safely 3. Follow Emergency Response Procedures 4. Maintain And Improve Health and Safety in the Work Place	10
SICIP-RMG-JP-02-G	Perform Basic IT Skills	1. Identify And Use Most Commonly Used IT Tools 2. Operate Computer. 3. Work With Word Processing Software. 4. Use Spread Sheet Packages to Create /Prepare Worksheets 5. Use Presentation Packages to Create / Prepare Presentation 6. Print The Documents 7. Use The Internet and Access E-Mail	10
SICIP-RMG-JP-03-G	Operate in a Self-Directed Team	1. Identify Team Goals and Processes 2. Communicate And Cooperate with Team Members 3. Work As a Team Member 4. Solve Problems as a Team Member	10
		Total Hour	**30**

Sector Specific Competencies (20 hrs.)

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP-RMG-JP-01-S	Perform Measurement and Calculations	1. Select Measuring Devices 2. Obtain Measurements for Apparel 3. Perform Simple Calculations	10
SICIP-RMG-JP-02-S	Interpret Sketch and Specifications in Manuals	1. Identify Information from Manual 2. Interpret Sketch and Specifications	10
		Total Hour	**20**

Occupation Specific Competencies (222 hrs.)

Code	Unit of Competency	Elements of Competency	Duration (hours)
SICIP-RMG-JP-01-O	Interpret Fundamentals of Knitting and Sweater	1. Recognize knitting and knitting types 2. Analyze knitting structure of sweater 3. Identify sweater and types 4. Interpret technical specifications of sweater 5. Illustrate the manufacturing process of sweater	16
SICIP-RMG-JP-02-O	Identify Yarn & Sweater Machinery	1. Identify types of yarns for sweater 2. Identify sweater knitting machinery 3. Recognize major parts of jacquard machine	12
SICIP-RMG-JP-03-O	Operate Jacquard Knitting Machine	1. Follow OHS practices 2. Prepare for knitting 3. Perform jacquard machine operation 4. Clean and maintain workplace	24
SICIP-RMG-JP-04-O	Practice Jacquard Programming	1. Prepare for jacquard programming 2. Interpret jacquard software interface 3. Use programming software tools 4. Exercise with freehand design	80
SICIP-RMG-JP-05-O	Develop Jacquard Program	1. Analyze design 2. Make swatch program 3. Perform program for sample 4. Finalize jacquard program	90
		Total Hour	**222**

COMPETENCY STANDARD: JACQUARD PROGRAMKING

The Generic Competencies

Unit of Competency: **APPLY OHS PRACTICES IN THE WORKPLACE**	**Nominal Duration:** 10 hrs.	**Unit Code:** SICIP-RMG-JP-01-G
Unit Descriptor: This unit covers knowledge, skills and attitudes required to apply occupational health and safety (OHS) practices in the workplace. It specifically includes the tasks of identifying, controlling and reporting OHS hazards, conducting work safely, following emergency response procedures, and maintaining and improving health and safety in the workplace.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	**Performance Criteria**
1. Identify, control and report OHS hazards	1.1 Immediate work area is routinely checked for OHS hazards prior to commencing and during work. 1.2 **<u>Hazards</u>** and unacceptable performance are identified and corrective action is taken within the level of responsibility. 1.3 OHS hazards and incidents are reported to appropriate personnel according to workplace procedures. 1.4 Safety Signs and symbols are identified and followed.
2. Conduct work safely	2.1. Apply OHS practices in the workplace. 2.2. Appropriate **<u>Personal Protective Equipment (PPE)</u>** is selected and worn.
3. Follow emergency response procedures	3.1 Emergency situations are identified and reported according to workplace reporting requirements. 3.2 Emergency procedures are followed as appropriate to the nature of the emergency and according to workplace procedures. 3.3 **<u>Workplace procedures</u>** for dealing with accidents, fires and emergencies are followed whenever necessary within scope of responsibilities.
4. Maintain and improve health and safety in the workplace	4.1 Risks are identified and appropriate control measures are implemented in the work area. 4.2 Recommendations arising from risk assessments are implemented within level of responsibility. 4.3 Opportunities for improving OHS performance are identified and raised with relevant personnel. 4.4 Safety records according to **<u>company policies</u>** are maintained.

Range of Variables

Variable	**Range** May include but not limited to:
1. Hazards	1.1 OHS incidents include near misses, injuries, illnesses and property damage, noise, handling hazardous substances, other hazards 1.2 Working with and near moving equipment/load shifting equipment 1.3 Broken or damaged equipment or materials
2. Personal protective equipment (PPE)	1.4 Goggles 1.5 Scarf 1.6 Gloves 1.7 Clothing 1.8 Apron 1.9 Safety shoes
3. Workplace procedures	1.10 OHS system and related documentation including policies and procedures 1.11 Standard Operating Procedures (SOPs) 1.12 Information on hazards and the work process, hazard alerts, safety signs and symbols 1.13 Labels 1.14 Material Safety Data Sheets (MSDSs) and manufacturers' advice
4. Company policies	4.1 Job-related Standard Operating Procedures (SOPs) and OHS-specific procedures. Examples of OHS procedures include consultation and participation, emergency response, response to specific hazards, incident investigation, risk assessment, reporting arrangements and issue resolution procedures.

Curricular Content Guide:

1. Underpinning Knowledge	1.1 Personal protective equipment (PPE) 1.2 Hazardous events 1.3 Tools, equipment, machinery and relevant accessories 1.4 Communication procedures 1.5 Job roles, responsibilities and compliance 1.6 Workplace laws 1.7 Company policies
2. Underpinning Skills	2.1 Using appropriate PPE 2.2 Identifying tools and equipment 2.3 Responding quickly and taking safety precautions for different hazardous situations 2.4 Operating and using tools, equipment, machinery and accessories properly as per SOP (Company standards) 2.5 Applying OHS practices in the workplace 2.6 Communicating with peers and supervisors 2.7 Applying workplace procedures for dealing with

	accidents, fires and emergencies 2.8 Maintaining safety records according to company policies
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Good relationships with peers, sub-ordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Tools, equipment and physical facilities appropriate to perform activities 4.3 Materials, consumables to perform activities 4.4 Personal protective equipment (PPE) 4.5 Emergency response manual

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 described workplace laws 1.2 explained company policies 1.3 used appropriate PPE 1.4 operated and used tools, equipment, machinery and accessories properly as per SOP (Company standards) 1.5 applied emergency response procedures 1.6 maintained and improved health and safety in the workplace
2. Methods of Assessment	Methods of assessment may include but not limited to: 2.1 Written test 2.2 Practical demonstration 2.3 Oral question 2.4 Portfolio (optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training 3.2 Assessment should be done by a certified assessor or occupation-specific industry expert.

<table>
<tr><td>Unit of Competency:
PERFORM BASIC IT SKILLS</td><td>Nominal Duration:
10 hrs.</td><td>Unit Code:
SICIP-RMG-JP-02-G</td></tr>
<tr><td colspan="3">Unit Descriptor:
This unit covers the knowledge, skills and attitudes required to perform basic IT skills. It specifically includes the tasks of identifying and using most commonly used IT tools, operating computer, working with word processing software, using spread sheet packages to create/prepare worksheets, using presentation packages to create/prepare presentation, printing the documents, and using the internet and accessing e-mail.</td></tr>
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Elements and Performance Criteria:
(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify and use most commonly used IT tools	1.1 Context of IT is interpreted. 1.2 Commonly used **IT tools** are identified. 1.3 Safe work practice and OHS Standards are followed.
2. Operate computer	2.1 **Peripherals** are checked and connected with computer as per standard. 2.2 Power cords/adapter are connected with computer and power outlets socket safely. 2.3 Computer is switched on gently. 2.4 PC **desktop/GUI** settings are arranged and customized as per requirement. 2.5 Files and folders are created, opened, copied, renamed, deleted and sorted as per requirement. 2.6 Properties of files and folders are viewed and searched. 2.7 Disks are defragmented, formatted as per requirement.
3. Work with word processing software	3.1 Word processing software is selected and started. 3.2 Basic typing technique is demonstrated. 3.3 **Documents** are created as per requirement in personal use and office environment. 3.4 **Contents** are entered. 3.5 Documents are **formatted**. 3.6 Paragraph and page settings are completed. 3.7 Saving and retrieving technique of a document are interpreted.
4. Use spread sheet packages to create/ prepare worksheets	4.1 Spread sheet packages are selected and started. 4.2 Worksheets are created as per requirement in Personal use and office environment. 4.3 Data are entered. 4.4 **Functions** are used for calculating and editing logical operation. 4.5 Sheets are formatted as per requirement. 4.6 Charts are created. 4.7 Charts/sheets are previewed.
5. Use presentation	5.1 Appropriate presentation software packages are selected

packages to create/prepare presentation	and started. 5.2 Presentation is created as per requirement in personal use and office environment. 5.3 Image, Illustrations, text, table, symbols and media are entered as per requirements. 5.4 Presentations are formatted and animated. 5.5 Presentations are previewed.
6. Print the documents	6.1 Printer is connected with computer and power outlet properly. 6.2 Power is switched on at both the power outlet and printer. 6.3 Printer is installed and added. 6.4 Correct printer settings are selected and document is printed.
7. Use the internet and access e-mail	7.1 Appropriate internet **browsers** are selected. 7.2 Search engines are used to access information. 7.3 Video/information are shared/downloaded/uploaded from/ to web site/social media. 7.4 Web based resources are used. 7.5 E-mail services are identified and selected to create a new e-mail address. 7.6 Document is prepared, attached and sent to different types of recipients. 7.7 e-mail is read, forwarded, replied and deleted as per requirement. 7.8 Custom e-mail folders are created and manipulated. 7.9 e-mail message is printed.

Range of Variables

Variable	Range May Include but not limited to:
1. IT tools	1.1 Phone 1.2 Cell Phone 1.3 TABs 1.4 Radio 1.5 Television 1.6 Computers 1.7 Laptops 1.8 Notebooks 1.9 Internet 1.10 Software 1.11 Satellite
2. Peripherals	2.1 Monitor 2.2 Keyboard 2.3 Mouse 2.4 Modem 2.5 Scanner 2.6 Printer
3. Desktop/GUI	3.1 Icons

settings	3.2 Taskbar 3.3 View 3.4 Resolutions
4. Documents	4.1 Word documents 4.2 Standard CV/bio-data with different text & fonts, image and table 4.3 Application/official letter with proper paragraph and indenting, spacing, styles, Illustrations, tables, header & footers and symbols 4.4 Standard report/newspaper items with column, footnote and endnote, drop cap, indexing and page numbering
5. Contents	5.1 Illustrations and styles 5.2 Text 5.3 Table 5.4 Symbols 5.5 Header & footer
6. Formatted	6.1 Bold 6.2 Italic 6.3 Underline 6.4 Font size, colour, 6.5 Change case 6.6 Alignment and intend
7. Functions	7.1 Mathematical 7.2 Logical 7.3 Simple statistical
8. Browsers	8.1 Internet explorer 8.2 Firefox 8.3 Google chrome 8.4 Opera 8.5 Safari 8.6 Omni web

Curricular Content Guide:

1. Underpinning Knowledge	1.1 Basic components of PC 1.2 IT and IT tools 1.3 Different types of software and application packages 1.4 Use of word processor, spread sheet and presentation software 1.5 Different types of mathematical, logical and conditional functions 1.6 Computer troubleshooting 1.7 Internet browsers and search engines 1.8 Techniques to access internet
2. Underpinning Skills	2.1 Identifying and using IT tools 2.2 Demonstrating simple troubleshooting with Computer 2.3 Demonstrating typing on word processing software 2.4 Saving and retrieving documents on word processing software

	2.5 Selecting and using appropriate presentation software packages for creating presentations 2.6 Demonstrating ability to create e-mail accounts 2.7 Opening an e-mail account and using it for different purposes 2.8 Configuring appropriate printer settings to print the documents 2.9 Using functions for calculating and editing logical operation in spread sheet
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Good relationships with peers, sub-ordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Learning manuals 4.3 Workplace equipment 4.4 IT Tools 4.5 Computers with word processing application 4.6 Internet connection

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 followed OHS standard and safe work procedures 1.2 created, opened, copied, renamed, deleted and sorted files and folders as per requirement 1.3 completed application software Installations properly 1.4 performed simple trouble shooting with Computer 1.5 demonstrated typing on word processing software, save and retrieve documents 1.6 used functions for calculating and editing logical operation in spread sheet 1.7 configured appropriate printer settings and printed the document 1.8 demonstrated ability to create and use e-mail accounts for different online purposes
2. Methods of Assessment	Methods of assessment may include but not limited to: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated

	work place after completion of the training 3.2 Assessment should be done by a certified assessor or occupation-specific industry expert

Unit of Competency: **OPERATE IN A SELF-DIRECTED TEAM**	Nominal Duration: 10 hrs.	Unit Code: SICIP-RMG-JP-03-G
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to operate in a self-directed team. It specifically includes the tasks of identifying team goals and processes, communicating and cooperating with team members, working and solving problems as a team member.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify team goals and processes	1.1 **Team goals** and processes are identified. 1.2 Roles and responsibilities of team members are identified. 1.3 Relationships within team and with other work areas are identified.
2. Communicate and cooperate with team members	2.1 Effective interpersonal skills are used to interact with team members and to contribute to activities and objectives. 2.2 Formal and informal forms of communication are used effectively to support team achievement. 2.3 Diversity is respected and valued in team functioning. 2.4 Views and opinions of other team members are understood and reflected accurately. 2.5 **Workplace staff regulation** is used correctly to assist communication.
3. Work as a team member	3.1 Duties, responsibilities, authorities, objectives and task requirements are identified and clarified with team. 3.2 Tasks are performed in accordance with organizational and team requirements, specifications and workplace procedures. 3.3 Team members support other members as required to ensure team achieves goals and requirements. 3.4 Agreed reporting lines are followed using standard operating procedures.
4. Solve problems as a team member	4.1 Current and potential problems faced by team are identified. 4.2 Procedures for avoiding and managing problems are identified. 4.3 Problems are solved effectively and in a manner that supports the team.

Range of Variables

Variable	Range May Include but not limited to:

1. Team goals and processes	1.1 Identification of the problem 1.2 Consideration of solutions 1.3 Action 1.4 Follow-up
2. Workplace staff regulation	2.1 Organization's/company's code of conduct, complaint handling/grievance policies and procedures

Curricular Content Guide:

1. Underpinning Knowledge	1.1 Organization requirements for written and electronic communication methods 1.2 Effective verbal communication methods 1.3 Roles and responsibilities of team members 1.4 Effective formal and informal forms of communication 1.5 Methods of identifying current and potential problems faced by a team 1.6 Effective problem-solving methods and evaluation of outcomes
2. Underpinning Skills	2.1 Identifying team goals and collaborative decision-making processes 2.2 Organizing information 2.3 Understanding and conveying intended meaning 2.4 Participating in a variety of workplace discussions 2.5 Compiling with Organization's requirements in the use of written and electronic communication methods 2.6 Performing tasks in accordance with organizational and team requirements, and workplace procedures and practice 2.7 Identifying current and potential problems faced by the team 2.8 Solving problems effectively and evaluating the outcome of the implemented solution
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Good relationships with peers, sub-ordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Work book 4.3 Learning manuals 4.4 Workplace tools/equipment 4.5 Communication tools

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 communicated and worked within a team in an interactive work environment as per workplace standard. 1.2 dealt with a range of communication/information at one time. 1.3 made constructive contributions in workplace issues 1.4 presented information clearly and effectively in written form 1.5 used effective interpersonal skills to interact with team members 1.6 solved problems as a team member
2. Methods of Assessment	Methods of assessment may include but not limited to: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training 3.2 Assessment should be done by a certified assessor or occupation-specific industry expert

The Sector Specific Competencies

Unit of Competency: **PERFORM MEASUREMENT AND CALCULATIONS**	**Nominal Duration:** 10 hrs.	**Unit Code:** SICIP-RMG-JP-01-S
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to perform measurement and calculations. It specifically includes the tasks of selecting measuring devices, obtaining measurements for apparel and performing simple calculations.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **<u>bold and underlined</u>** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Select measuring devices	1.1 Work instructions are confirmed and applied to the job in hand. 1.2 Materials to be measured are identified as per job specifications. 1.3 Appropriate **<u>measuring devices</u>** are selected based on materials to be measured. 1.4 Specifications are obtained from relevant **<u>documents.</u>** 1.5 Tolerance and clearance limits are identified and adjusted according to the job requirements.
2. Obtain measurements for apparel	2.1 Accurate **<u>measurements</u>** are obtained in accordance with job requirements. 2.2 Systems of measurements are identified and measurement conversions done as per requirement. 2.3 Measurements are confirmed and recorded in the given company format.
3. Perform simple calculations	3.1 Simple calculations involving **<u>basic operations</u>** are carried out. 3.2 **<u>Other operations</u>** are used to complete tasks. 3.3 Appropriate formulas for calculating quantities of materials are selected. 3.4 **<u>Calculations</u>** are performed and verified. 3.5 Material quantities are calculated and shared with team as per requirement.

Range of Variables

Variable	**Range** May include but not limited to:
1. Measuring device	1.1. Measuring Tape 1.2. Steel rule 1.3. Calculator 1.4. Sets square
2. Documents	2.1 Technical Manuals 2.2 Specifications

	2.3 Sketches 2.4 Charts 2.5 Photographs
3. Measurements	3.1 Length 3.2 Width 3.3 Weight 3.4 Tolerance
4. Basic operations	4.1 Addition 4.2 Subtraction 4.3 Multiplication 4.4 Division
5. Other operations	5.1 Fractions 5.2 Percentages 5.3 Mixed numbers 5.4 Conversions 5.5 Scales
6. Calculations	6.1 Area 6.2 Volume 6.3 Circumference 6.4 CBM 6.5 Volumetric Weight

Curricular Content Guide

1. Underpinning Knowledge	1.1 Information on measuring devices 1.2 Selection technique of appropriate measuring devices 1.3 Measurement and calculation techniques for apparel merchandising 1.4 Instructions for using measuring devices
2. Underpinning Skills	2.1 Identifying measuring devices based on materials to be measured 2.2 Obtaining specification of measuring devices from relevant documents 2.3 Taking measurement according to job requirements 2.4 Identifying tolerance and clearance limits and adjusting according to the job requirements 2.5 Performing calculations for quantities of materials 2.6 Confirming and recording measurements as per standard
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Good relationships with peers, sub-ordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided:

	4.1 Workplace (simulated or actual) 4.2 Measurement tools/devices 4.3 Technical manuals 4.4 Specifications, sketches, charts, photographs

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 selected measuring devices based on materials to be measured 1.2 obtained measurements as per job requirements 1.3 performed calculations for quantities of materials 1.4 confirmed and recorded measurements as per standard
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **INTERPRET SKETCH AND SPECIFICATIONS IN MANUALS**	**Nominal Duration:** 10 hrs.	**Unit Code:** SICIP-RMG-JP-02-S
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to interpret sketch and specifications in manuals. It specifically includes the tasks of identifying information from manual and interpreting sketch and specifications.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify information from manual	1.1 Appropriate **manuals** are collected as per sample. 1.2 Importance of manuals is recognized. 1.3 Required information are collected from manuals.
2. Interpret sketch and specifications	2.1 Relevant **sketches** and **specifications** are identified. 2.2 Key **terms and abbreviations** are identified. 2.3 **Signs and symbols** are identified. 2.4 Schedules, dimensions, drawings and specifications are interpreted.

Range of Variables

Variable	Range May include but not limited to:
1. Manuals	1.1 Buyers' specification manual 1.2 Compliance manual 1.3 Maintenance procedure manual 1.4 Periodic maintenance manual 1.5 Quality manual 1.6 Signs and symbols, instruction manuals
2. Sketches	2.1 Technical sketch 2.2 Measurement sketch
3. Specifications	3.1 Product specifications 3.2 Performance specifications 3.3 Method specifications
4. Terms and abbreviations	4.1 Refers to all terms and abbreviations associated with the RMG sector
5. Signs and symbols	5.1 Include all signs and symbols associated with the RMG sector

Curricular Content Guide

1. Underpinning Knowledge	1.1 Themes on various types of RMG manuals 1.2 Rules of sketch, drawings and specifications 1.3 Schedules, dimensions, drawings and specifications
2. Underpinning Skills	2.1 Identifying relevant sketches and specifications

	2.2 Identifying key terms, abbreviations, signs and symbols 2.3 Selecting and collecting appropriate manuals as per sample 2.4 Collecting required information from manual
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Good relationships with peers, sub-ordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Manuals 4.3 Sketches, drawings, specifications

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 collected information from manual as per sample 1.2 identified sketches and specifications as per sample 1.3 interpreted schedules, dimensions, drawings and specifications
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

The Occupation Specific Competencies

Unit of Competency: **INTERPRET FUNDAMENTALS OF KNITTING AND SWEATER**	Nominal Duration: 16 hrs.	Unit Code: SICIP-RMG-JP-01-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to interpret fundamentals of knitting and sweater. It specifically includes the tasks of recognizing knitting and knitting types, analyzing knitting structure of sweater, identifying sweater and types, interpreting technical specifications of sweater and illustrating the manufacturing process of sweater.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Recognize knitting and knitting types	1.1. Knitting is defined as the process of interloping yarns or machine knitting. 1.2. **Types of knitting** are recognized and described based on their characteristics and uses. 1.3. Advantages and disadvantages of different knitting types are explained.
2. Analyze knitting structure of sweater	2.1 **Knitting structures of the sweater** is recognized based on fabric characteristics. 2.2 Sample of the sweater is identified and analyzed.
3. Identify sweater and types	3.1 **Types of sweaters** are identified according to styling features such as neckline, fit and closure method. 3.2 **Gauge of sweater** is measured as per the buyer's standard using appropriate tools and procedures. 3.3 Style differences between sweaters are recognized based on visual and construction features.
4. Interpret technical specifications of sweater	4.1 **The technical package of the sweater** is interpreted. 4.2 **Parts of the sweater** and labels are identified. 4.3 **Points of measurement** for sweater dimensions are identified and understood. 4.4 **Embellishments** and decorative elements are identified and described according to the design specifications.
5. Illustrate the manufacturing process of sweater	5.1 **Sweater manufacturing processes** are identified and explained. 5.2 **Sections in sweater manufacturing** are listed and described. 5.3 Functions of different sections in the sweater manufacturing process are described. 5.4 Quality control measures are explained and identified at various stages of the manufacturing process.

Range of Variables

Variable	**Range** May include but not limited to:
1. Types of knitting	1.1 Based on fabric manufacturing 1.1.1 Circular knit 1.1.2 Flat knit 1.2 Based on yarn filling 1.2.1 Warp knitting 1.2.2 Weft knitting
2. Knitting structures of sweater	2.1 Rib 2.2 Jersey 2.3 Reverse jersey 2.4 Half cardigan 2.5 Full cardigan 2.6 Pointel stitch 2.7 Pearl stitch 2.8 Tubular 2.9 Cable 2.10 Special structure: 2.10.1 Intarsia knit 2.10.2 Structure knit 2.11 Jacquard knit
3. Types of sweaters	3.1 Pull over 3.1.1 Round neck 3.1.2 V-neck 3.1.3 Turtleneck 3.1.4 Mock neck 3.1.5 Vest type 3.1.6 Polo type 3.2 Cardigan 3.2.1 Open cardigan 3.2.2 Zipper cardigan 3.3 Button cardigan
4. Gauge of sweater	4.1 Heavy weight 4.1.1 1.5 Gauge 4.1.2 3 Gauge 4.1.3 5 Gauge 4.1.4 7 Gauge 4.2 Light weight 4.2.1 9 Gauge 4.2.2 12 Gauge 4.2.3 14 Gauge 4.3 16 Gauge
5. The technical package of the sweater	5.1 Sketch/Photograph 5.2 Yarn specification 5.3 Knitting construction details 5.4 Measurement chart

	5.5 Trims and accessories details 5.6 Embellishment artwork 5.7 Washing instruction 5.8 Finishing and packing instruction
6. Parts of the sweater	6.1 Tops 6.1.1 Front part 6.1.2 Back part 6.1.3 Sleeve 6.1.4 Neck 6.1.5 Hood 6.1.6 Cuff 6.1.7 Pocket 6.1.8 Placket 6.1.9 Elbow 6.1.10 Moon 6.1.11 Rib 6.1.12 Neck tape/shoulder tape 6.1.13 Piping 6.1.14 Zipper 6.2 Bottom 6.2.1 Front panel 6.2.2 Back panel 6.2.3 Waist bend 6.2.4 Pocket 6.2.5 Drawstring 6.2.6 Placket 6.3 Rib
7. Points of measurements	7.1 Tops 7.1.1 High Point Shoulder (HPS) 7.1.2 Rib height 7.1.3 Bottom opening 7.1.4 Shoulder length 7.1.5 Full length 7.1.6 Overarm sleeve length 7.1.7 Underarm sleeve length 7.1.8 Centre front length 7.1.9 Centre back length 7.1.10 Chest 7.1.11 Across front 7.1.12 Across back 7.1.13 Neck width 7.1.14 Neck depth 7.1.15 Waist 7.1.16 Arm hole 7.1.17 Sleeve open 7.2 Bottom 7.2.1 Full length 7.2.2 Hip/seat

	7.2.3 Inseam 7.2.4 Waist 7.2.5 Thigh 7.2.6 Front rise 7.2.7 Back rise 7.2.8 Knee 7.2.9 Leg opening 7.2.10 Pocket length 7.3 Pocket width
8. Embellishment	8.1 Embroidery 8.2 Print 8.3 Wash 8.4 Piece Dye
9. Sweater manufacturing processes	9.1 In house of yarn 9.2 Winding 9.3 Knitting 9.4 Inspection 9.5 Linking 9.6 Trimming 9.7 Light check 9.8 Mending 9.9 Sewing 9.10 Washing 9.11 Attachment 9.12 Ironing 9.13 Quality check 9.14 Packing
10. Sections in sweater manufacturing	10.1 Production Sections 10.1.1 Winding 10.1.2 Programming 10.1.3 Knitting 10.1.4 Linking 10.1.5 Trimming 10.1.6 Mending 10.1.7 Washing 10.1.8 Sewing 10.1.9 Finishing 10.1.10 Packing 10.2 Cross Functional Sections 10.2.1 Merchandising 10.2.2 Sample 10.2.3 Planning 10.2.4 Store 10.2.5 Quality Assurance 10.2.6 Industrial Engineering (IE) 10.2.7 Maintenance 10.2.8 Human Resource (HR) 10.3 Compliance

Curricular Content Guide

1. Underpinning knowledge	1.1 Definition of knitting and sweater 1.2 Knitting types and structures 1.3 Types of sweater and accessories 1.4 Weight of sweater 1.5 Technical package of sweater 1.6 Parts, points and measurements 1.7 Embellishment 1.8 Sweater manufacturing process 1.9 Departments of sweater factory
2. Underpinning skills	2.1 Recognizing knitting, sweater and types 2.2 Identifying knitting structures 2.3 Identifying sweater and accessories 2.4 Identifying parts, points and measurements 2.5 Identifying Embellishments and decorative elements according to the design specifications 2.6 Identifying sweater manufacturing processes 2.7 Identifying quality control measures at various stages of the manufacturing process
3. Underpinning attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Good relationships with peers, sub-ordinates and seniors in the workplace
4. Resource implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Sweater samples 4.3 Technical package 4.4 Measuring tape 4.5 Books on sweaters 4.6 Accessibility to the workplace 4.7 Learning/operational manual

Assessment Evidence Guide

1. Critical aspects of competency	Assessment must evidence that the candidate: 1.1 recognized knitting and knitting types 1.2 analyzed knitting structure of sweater 1.3 interpreted technical specifications of sweater 1.4 identified quality control measures at various stages of the manufacturing process 1.5 illustrated the manufacturing process of sweater 1.6 identified embellishments and decorative elements
2. Methods of	Methods of assessment may include but is not limited to:

assessment	2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (optional)
3. Context of assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **IDENTIFY YARN AND SWEATER MACHINERY**	Nominal Duration: 12 hrs.	Unit Code: SICIP-RMG-JP-02-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to identify yarn and sweater machinery. It specifically includes the tasks of identifying types of yarns for sweater, identifying sweater knitting machinery and recognizing major parts of jacquard machine.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify types of yarns for sweater	1.1. **Types of yarns** are identified and classified according to their fiber content. 1.2. **Yarn count** is interpreted to determine the thickness or weight of the yarn. 1.3. **Common yarn counts** are identified based on the intended use of the sweater. 1.4. Characteristics of yarns are described to understand their suitability for sweater production. 1.5. Color and shade codes of yarn are recognized and verified according to the manufacturer's specifications.
2. Identify sweater knitting machinery	2.1. **Sweater knitting machine** is identified and described according to its type. 2.2. **Gauge (GG)** of the sweater knitting machine is interpreted. 2.3. Features of jacquard knitting machines are described. 2.4. Operating principles of the sweater knitting machine are explained.
3. Recognize major parts of jacquard machine	3.1 **Jacquard machines** are identified as per brand and country of origin. 3.2 **Major parts** of jacquard machine are identified. 3.3 Functions of major parts are interpreted.

Range of Variables

Variable	**Range** May include but not limited to:
1. Types of yarns	1.1 Cotton 1.2 Acrylic 1.3 Nylon 1.4 Rayon 1.5 Polybutylene Terephthalate (PBT) 1.6 Elastic 1.7 Spandex 1.8 Polyester

	1.9 Wool 1.9.1 Lamb's 1.9.2 Marino 1.10 Mohair 1.11 Blended 1.12 Fancy
2. Yarn count	2.1 Direct system 2.2 Indirect system
3. Common yarn counts	3.1 Singles 3.2 Doubles 3.2.1 Cotton 3.2.1.1 16/2 3.2.1.2 40/2 3.2.1.3 32/2 3.2.1.4 20/2 3.2.2 Acrylic 3.2.2.1 36/2 3.2.2.2 32/2 3.2.2.3 28/2
4. Sweater knitting machine	4.1 Manual knitting machine 4.2 Semi auto machine 4.3 Automated/Jacquard knitting machine 4.4 Whole garments/advanced knitting machine
5. Gauge (GG)	5.1 1.5 GG 5.2 GG 5.3 GG 5.4 7 GG 5.5 9 GG 5.6 12 GG 5.7 14 GG 5.8 16 GG
6. Jacquard Machines	6.1 Stoll (Germany) 6.2 Shima Seiki (Japan) 6.3 Different Chinese brands
7. Major parts	7.1 Power switch 7.2 Emergency switch 7.3 Monitor 7.4 CPU unit 7.5 System carriage 7.6 Machine bed 7.7 Needle 7.8 Knitting cam 7.9 Sinker 7.10 Jack 7.11 Feeder 7.12 Feeder rail 7.13 Starting bar 7.14 Side tension

	7.15 Yarn roller 7.16 Top tension spring 7.17 Alarm signal 7.18 Comb guide 7.19 Main roller 7.20 Clamping bed/Gripper 7.21 Carriage/Head

Curricular Content Guide

1. Underpinning knowledge	1.1 Sweater yarn and yarn count 1.2 Color and codes of sweater yarn 1.3 Sweater knitting machine 1.4 Machine gauge of sweater knitting machine 1.5 Jacquard knitting machine and its major parts
2. Underpinning skills	2.1 Identifying sweater yarn, types and count 2.2 Identifying sweater knitting machine and gauge 2.3 Identifying jacquard knitting machine and major parts
3. Underpinning attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Good relationships with peers, sub-ordinates and seniors in the workplace
4. Resource implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Yarn samples 4.3 Sweater knitting machine 4.4 Accessibility to the workplace 4.5 Learning/operational manual

Assessment Evidence Guide

1. Critical aspects of competency	Assessment must evidence that the candidate: 1.1 described color and codes of sweater yarn 1.2 identified types of yarns and counts 1.3 identified sweater knitting machinery 1.4 recognized jacquard major parts of machine
2. Methods of assessment	Methods of assessment may include but is not limited to: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (optional)
3. Context of assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or

		simulated work place after completion of the training.
	3.2	Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **OPERATE JACQUARD KNITTING MACHINE**	Nominal Duration: 24 hrs.	Unit Code: SICIP-RMG-JP-03-O
Unit Descriptor: This unit of competency covers the knowledge, skills and attitudes required to operate jacquard knitting machine. It specifically includes the tasks of following OHS practices, preparing for knitting, performing jacquard machine operation, and cleaning and maintaining workplace.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Follow OHS practices	1.1 **Personal protective equipment (PPE)** is collected, inspected, and worn according to job requirements. 1.2 Occupational health and safety (OHS) protocols are followed. 1.3 Safety measures and safe material handling are adhered to. 1.4 Safety checks are conducted to ensure compliance with OHS standards.
2. Prepare for knitting	2.1 Machine cleaning and checks are performed. 2.2 Machine is switched on according to manufacturer's manual. 2.3 Program for the sweater part is loaded into the machine according to the specific design. 2.4 Yarn setting is carried out based on the knitting requirements. 2.5 Carrier/feeder and tension values are set to ensure proper yarn feeding. 2.6 Take down speed and knitting speed are adjusted as per the knitting requirements.
3. Perform jacquard machine operation	3.1 Machine is run according to the operational procedures. 3.2 Front part of the sweater is knitted as per the styling specifications. 3.3 Back part of the sweater is knitted according to the styling requirements. 3.4 Sleeve is knitted as per the styling design, ensuring proper fit and pattern alignment. 3.5 Various types of necks are knitted as per styling, ensuring that the neckline is appropriately designed. 3.6 **Troubleshooting** is carried out in case of any problems. 3.7 Knitted parts are checked to ensure **quality** by inspecting the stitches and patterns. 3.8 Machine is switched off properly after operation.

4. Clean and Maintain Workplace	4.1 Waste materials are disposed of following workplace standard operating procedures (SOPs). 4.2 Machines are cleaned ensuring all parts are free from dust, debris and residue. 4.3 Workplace is cleaned and organized according to workplace requirements.

Range of Variables

Variable	**Range** May include but not limited to:
1. Personal protective equipment (PPE)	1.1 Mask 1.2 Apron 1.3 Scarf/cap 1.1 Ear plug
2. Troubleshooting	2.1 Problematic needles change 2.2 Tension and speed adjustments 2.2.1 Re-establish yarn feeding
3. Quality	3.1 Measurements 3.2 Defects 3.2.1 Hole 3.2.2 Needle Missing 3.2.3 Oil Mark 3.4 Tension 3.5 Design

Curricular Content Guide

1. Underpinning knowledge	1.1 Personal protective equipment (PPE) 1.2 Occupational health and safety (OHS) standard 1.3 Jacquard machine operation 1.4 Yarn setting 1.5 Carrier, tension, take down speed and knitting speed 1.6 Standard operating procedures (SOPs)
2. Underpinning skills	2.1 Applying OHS practices in the workplace 2.2 Collecting, inspecting and wearing PPE 2.3 Loading program in the jacquard knitting machine and performing its operation 2.4 Setting yarn in the machine 2.5 Setting carrier, tension, take down speed and knitting speed 2.6 Cleaning jacquard knitting machine 2.7 Cleaning and organizing workplace according to workplace requirements
3. Underpinning attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness

	3.7 Respect for rights of peers and seniors in the workplace 3.8 Good relationships with peers, sub-ordinates and seniors in the workplace
4. Resource implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 PPE 4.3 Yarns 4.4 Sweater part 4.5 Program software 4.6 Jacquard knitting machine 4.7 Accessibility to the workplace 4.8 Learning manual

Assessment Evidence Guide

1. Critical aspects of competency	Assessment must evidence that the candidate: 1.1 described jacquard machine operational procedure 1.2 applied OHS practices 1.3 set carrier, tension, take down speed and knitting speed 1.4 loaded program in the jacquard knitting machine and performed its operation 1.5 cleaned and maintained jacquard knitting machine and workplace
2. Methods of assessment	Methods of assessment may include but is not limited to: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (optional)
3. Context of assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **PRACTICE JACQUARD PROGRAMMING**	Nominal Duration: 80 hrs.	Unit Code: SICIP-RMG-JP-04-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to practice jacquard programming. It specifically includes the tasks of preparing for jacquard programming, interpreting jacquard software interface, using programming software tools and exercising with freehand design.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Prepare for jacquard programming	1.1 Technical package or sample is selected and collected for Jacquard programming. 1.2 **<u>Computer hardware and peripherals</u>** are checked for proper functioning. 1.3 Appropriate **<u>software</u>** is identified based on the requirements of the Jacquard machine. 1.4 Software is opened according to the manufacturing standards. 1.5 Pre-programming checks are performed to ensure all settings are configured.
2. Interpret jacquard software interface	2.1 Jacquard software interface is interpreted and understood. 2.2 Menu bar of the software is identified. 2.3 **<u>Tabs and icons</u>** within the software are identified and categorized based on their functionality. 2.4 Function of icons is described and explained their specific role in the software.
3. Use programming software tools	3.1 New pattern is inserted into the system according to the design specifications. 3.2 **<u>Programming software tools</u>** are identified and understood for their role in the programming process. 3.3 Function of tools is interpreted. 3.4 Tools are used to create programs by inputting design data. 3.5 Programming tools are used to create accurate, fast and high-quality jacquard patterns.
4. Exercise with freehand designs	4.1 **<u>Knitting chart</u>** is selected and collected. 4.2 Programs are practiced according to the knitting chart. 4.3 **<u>Controlling parameters</u>** are adjusted according to the software and machine requirements. 4.4 Draft program is developed and checked. 4.5 Draft program is compiled with error check by performing a test run with the design is knitted.

Range of Variables

Variable	**Range** May include but not limited to:
1. Computer hardware and peripherals	1.1. CPU 1.2. Monitor 1.3. Keyboard 1.4. Mouse 1.5. Scanner 1.6. Pen drive 1.7. Printer
2. Software	2.1 M 1 Plus (Stoll) 2.2 SDS One Apex 3/4 (Shima Seiki) 2.3 HQPDS (China) 2.4 Reynen (China) 2.5 CKS (China)
3. Tabs and icons	3.1 Select machine type 3.2 Save with pattern name 3.3 Needle 3.4 Courses 3.5 Basic techniques 3.6 Design 3.7 Repetition 3.8 Feeder setting 3.9 Insert commands and techniques 3.10 Check programs 3.11 Processing 3.12 Saving
4. Programming software tools	4.1 Selection 4.2 Insert row 4.3 Insert column 4.4 Delete row 4.5 Delete column 4.6 Mirror 4.7 Shapes 4.8 Moving 4.9 Fill copy 4.10 Batch copy 4.11 Matrix tools 4.12 Rectangle tools 4.13 Replace tools 4.14 Picture processing 4.15 Shield color tools
5. Knitting chart	5.1 Style 5.2 Measurement 5.3 Gauge 5.4 Yarn type

	5.5 Yarn count 5.6 Yarn composition 5.7 Weight 5.8 Needles and courses
6. Controlling Parameters	6.1 All needle 6.2 Rib category 6.3 Rib course 6.4 Body course 6.5 Slopping course 6.6 Shoulder hole 6.7 Neck drop hole 6.8 Sleeve mark

Curricular Content Guide

1. Underpinning knowledge	1.1 Computer hardware and peripherals 1.2 Menu bar, tabs, icons and tools of jacquard software 1.3 Jacquard software interface 1.4 Manual knitting chart 1.5 Necessary values for knitting
2. Underpinning skills	2.1 Selecting and collecting technical package or sample for Jacquard programming 2.2 Identifying appropriate software based on the requirements of the jacquard machine 2.3 Identifying menu bar, tabs, icons and tools of jacquard software 2.4 Identifying programming software tools for their role in the programming process 2.5 Selecting and collecting manual knitting chart 2.6 Adjusting controlling parameters according to the software and machine requirements 2.7 Practicing program 2.8 Inputting necessary values in the software 2.9 Developing final program
3. Underpinning attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in workplace 3.8 Good relationships with peers, sub-ordinates and seniors in workplace
4. Resource implications	The Following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Job specifications and work instructions 4.3 Computer and related accessories including internet facility 4.4 Jacquard software

	4.5 Manual knitting chart 4.6 Accessibility to the workplace 4.7 Learning/operating manual

Assessment Evidence Guide

1. Critical aspects of competency	Assessment must evidence that the candidate: 1.1 described computer hardware and peripherals 1.2 interpreted jacquard software interface 1.3 identified appropriate software based on the requirements of the jacquard machine 1.4 practiced program 1.5 inputted necessary values in the software 1.6 developed the final program
2. Methods of assessment	Methods of assessment may include but is not limited to: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (optional)
3. Context of assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **DEVELOP JACQUARD PROGRAM**	Nominal Duration: 90 hrs.	Unit Code: SICIP-RMG-JP-05-O
Unit Descriptor: This unit of competency covers the knowledge, skills and attitudes required to develop jacquard program. It specifically includes the tasks of analyzing design, making swatch program, performing program for sample and finalizing jacquard program.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Analyze Design	1.1 **<u>Forms of designs</u>** are interpreted. 1.2 Designs are collected from **<u>appropriate sources</u>**. 1.3 Designs are analyzed to assess elements. 1.4 Impact of design on production processes, including feasibility and cost considerations, is evaluated. 1.5 Design trends and market preferences are researched and incorporated.
2. Make Swatch Program	2.1 Appropriate machine model is selected based on the type of sweater part. 2.2 Loops are arranged as per the knitting structure. 2.3 Motifs are imported into the sample program according to the design requirements. 2.4 Panel is prepared as per the specified shape. 2.5 **<u>Programming parameters</u>** are set and adjusted to ensure pattern quality. 2.6 Processing of the program for the Jacquard machine is performed as per the specific requirements. 2.7 Swatch is prepared to view the visual outlook. 2.8 Program is saved according to workplace standards.
3. Perform program for sample	3.1 The appropriate machine model is selected based on the type of sweater part and its specific knitting requirements. 3.2 Loops are arranged according to the knitting structure to ensure consistency and correct pattern alignment. 3.3 Motifs are imported into the sample program as per the design specifications to match the desired pattern. 3.4 The panel is prepared to the specified shape, ensuring the accurate dimensions for the sweater part. 3.5 Programming parameters are set and adjusted to optimize pattern quality, including stitch density and speed. 3.6 The program for the Jacquard machine is processed, ensuring it aligns with design and operational requirements. 3.7 The sample is developed according to the knitting chart

	and inspected for visual accuracy and quality. 3.8 The program is saved following workplace standards for future use and modifications.
4. Finalize Jacquard Program	4.1 Knitting information is inserted into the software, and the final program is compiled for use. 4.2 Program errors are adjusted as needed to ensure proper machine function. 4.3 Fabric view is checked, and 3D view is prepared and validated if the feature is available. 4.4 The Jacquard knitting machine is selected, and the program is loaded onto the machine. 4.5 Trial production is performed, and the sample is checked for accuracy. 4.6 The final program is adjusted if necessary and then released for bulk production.

Range of Variables

Variable	**Range** May include but not limited to:
1. Forms of design	1.1 Technical package 1.2 Photograph/image 1.3 Physical sample 1.4 Stitch swatch
2. Appropriate sources	2.1 Department head 2.2 Chief programmer 2.3 Merchandising department 2.4 Research and development (R&D) department
3. Programming parameters	3.1 All needle 3.2 Rib category 3.3 Rib course 3.4 Body course 3.5 Slopping course 3.6 Shoulder hole 3.7 Neck drop hole 3.8 Sleeve mark

Curricular Content Guide

1. Underpinning Knowledge	1.1 Forms of designs 1.2 Appropriate sources of design collection 1.3 Programming parameters of jacquard knitting machine 1.4 Fabric view 1.5 Swatch preparation 1.6 Knitting chart 1.7 Program loading process 1.8 Trial and bulk production
2. Underpinning Skills	2.1 Collecting and analyzing design 2.2 Arranging loops as per the knitting structure 2.3 Setting and adjusting necessary parameters to ensure

	pattern quality 2.4 Checking fabric view 2.5 Selecting jacquard machine for sample making 2.6 Preparing manual knitting chart 2.7 Preparing final program 2.8 Checking fabric view 2.9 Loading program in the jacquard knitting machine
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Good relationships with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Designs 4.3 Computer and related accessories with internet facility 4.4 Jacquard software 4.5 Jacquard knitting machine 4.6 Learning/operational manual 4.7 Manual knitting chart 4.8 Accessibility to the workplace 4.9 Drawings, specifications and work instructions

Assessment Evidence Guide

1. Critical aspects of competency	Assessment must evidence that the candidate: 1.1 interpreted forms of designs 1.2 set programming parameters 1.3 prepared swatch to view the visual outlook 1.4 loaded program onto the jacquard knitting machine 1.5 developed sample and knitting chart 1.6 completed jacquard program
2. Methods of assessment	Methods of assessment may include but is not limited to: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (optional)
3. Context of assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

END OF COMPETENCY STANDARD

Workshop/Lab Facility Standard

Course Name:	Jacquard Programming
Number of Trainees:	20

Course-wise Training Space (Theoretical Classroom, Workshop/ Lab/ Classroom cum Workshop):

- Classroom – 350 sft (33 sqm)
- Workshop/ lab – 800 sft (75 sqm)

  OR
- Classroom cum workshop – 1000 sft (93 sqm)

Major Training Equipment and Training Facilities:

S.N.	Major Equipment and Training facilities	Required facilities
1.	Computer/Laptops	01
2.	Multimedia projector & screen	01
3.	Internet connectivity	01
4.	Scanner	01
5.	Printer	01
6.	White board	01
7.	Jacquard Knitting Machines	05
8.	Desktop Computers (with software)	20
9.	Different Yarn Sample	20
10.	Sweater Sample	30
11.	Different type of knitting chart (Printed copy)	40

The following conditions must be fulfilled –

- The institute shall not use the same facilities for any other projects/organizations offering a similar course.
- The institute must provide sufficient evidence to prove ownership of the proposed training equipment.

The list denotes the minimum training equipment and facility required to effectively conduct training for a specific course. Additionally, the institute must ensure that all other necessary training tools, equipment, and furniture are available to meet the requirement of competency standards (CS) provided by SICIP.

For the operation of training course on Jacquard Programming, the institute must ensure the availability of at least 80% of the major training equipment and training facilities (according to the CS) to be eligible for SICIP training delivery. If the score is below 80%, the remaining equipment and facilities need to be installed before the commencement of the training.

The institute will also provide all other hand tools and power tools as per CS for 20 trainees. Also, they will arrange adequate seating arrangement and classroom setup for the 20 trainees.